MIL-H-38471B(USAF)
4 October 1972
SUPERSEDING
MIL-H-38471A(USAF)
4 October 1966

MILITARY SPECIFICATION

HAT AND INSECT NET HGU-18/P

1. SCOPE

1.1 This specification covers one type of hat made from cotton and nylon material with a removable mosquito net (headnet). The hat is designated HGU-18/P and is furnished in the following sizes as specified (see 6.2):

Schedule of Sizes

6-3/8	7-1/4
6-1/2	7-3/8
6-5/8	7-1/2
6-3/4	7-5/8
6-7/8	7-3/4
7	7-7/8
7-1/8	

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-375	Plastic Film, Flexible, Vinyl Chloride
V-L-61	Laces, Nylon
V-T-285	Thread, Polyester
KK-L-271	Leather, Cattlehide, Strap, Vegetable-Tanned
QQ-B-613	Brass, Leaded, And Non-Leaded: Flat Products (Plate, Bar, Sheet And Strip)
UU-P-268	Paper, Kraft, Untreated, Wrapping
PPP-B-636	Box, Fiberboard
PPP-T-60	Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging
DDD-L-20	Label, For Clothing, Equipage, And Tentage, (General Use)

Military

MIL-F-495	Finish, Chemical, Black, For Copper Alloys
MIL-C-3924	Cloth, Oxford, Cotton Warp And Nylon Filling, Quarpel Treated

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MIL-F-10884	Fasteners, Snap
MIL-C-38469	Cloth, Plain Weave, Nylon
MIL-C-43352	Cloth, Netting, Nylon, Tulle, Tricot

STANDARDS

Federal

FED-STD-191	Textile Test Methods
FED-STD-751	Stitches, Seams, And Stitchings

Military

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage

DRAWING

Air Force

66K1622	Hat And Insect Net HGU-18/P, Patterns

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Standard sample. Samples, when furnished, are solely for guidance and information to the supplier (see 6.3). Variation from the specification may appear in the sample, in which case the specification shall govern.

3.2 Materials.

3.2.1 Basic material. The basic material shall be olive green shade No. 107 (see 6.3) and shall conform to class 2 of MIL-C-3924.

3.2.2 Lining and interlining. The material for lining the crown and for interlining the brim shall conform to MIL-C-38469.

3.2.3 Netting. The material for the headnet shall conform to MIL-C-43352.

* 3.2.4 Elastic nylon braid. The elastic nylon braid shall be flat, 3/16 (plus or minus 1/32) inch wide. It shall be braided using 210-denier nylon dyed olive green 106, on 17 carriers, 1 end per carrier with not less than 96 picks braiding in a one over one under manner. There shall be 8 strands of 70-gage compounded natural rubber, synthetic rubber, or a mixture of both.

* 3.2.4.1 Colorfastness dyed braid. The dyed braid shall show fastness to laundering and light equal to or better than the standard sample. When no standard sample is available, the dyed braid shall show good fastness to laundering and light when tested as specified in 4.2.1.

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3.2.5 Chin strap. The chin strap shall be made from laces that are olive green shade No. 106 (see 6.3) and that conform to type II, class 2 of V-L-61, except that the requirements for length, tips, and tipping shall not apply.

3.2.6 Binding for outside edge of brim interlining. The binding for the outside edge of the brim interlining shall be made from 0.008-inch-thick plastic film conforming to type I, class 1 of L-P-375, except that the requirements for ultimate elongation transdirection, tear resistance, Elmendorf, pinholes per square yard, resistance to weather, cold crack, lacquer lifting, crocking, and bondability shall not apply. The film may be finished in a matte or embossed simulated fabric weave. The binding shall be cut 1/2 to 9/16 inch wide.

3.2.7 Thread. The thread for seaming and stitching shall be dyed shade S-1, Cable No. 66022 and shall conform to type I or III, class 1, size A and B of V-T-285. The dyed thread shall show good colorfastness to light and laundering.

3.2.8 Keeper. The keeper for the chin strap shall be made of natural color leather that is 6/64 to 8/64 inch thick and that conforms to type I, II, III, or IV, class 1, sub-class a, selection A, B, or C of KK-L-271 except that the requirements for treatment for fungicide paranitrophenol, crocking, cracking, breaking force, elongation, chloroform extract, water soluble material, total ash, and degree of tannage shall not apply.

3.2.9 Ventilators. The ventilators and the washers shall be brass conforming to composition 2 of QQ-B-613, finished with a black chemical finish conforming to MIL-F-495. The washers shall conform to the requirements of Figure 1, and the ventilators shall conform to the requirements of Figure 2 when tested as specified in 4.2.1.

* 3.2.10 Snap fastener. The snap fastener for the brim turn-up shall be a pronged ring head type conforming to type 1B, 3B, 4B and 5B, style 3, finish 2 of MIL-F-10884.

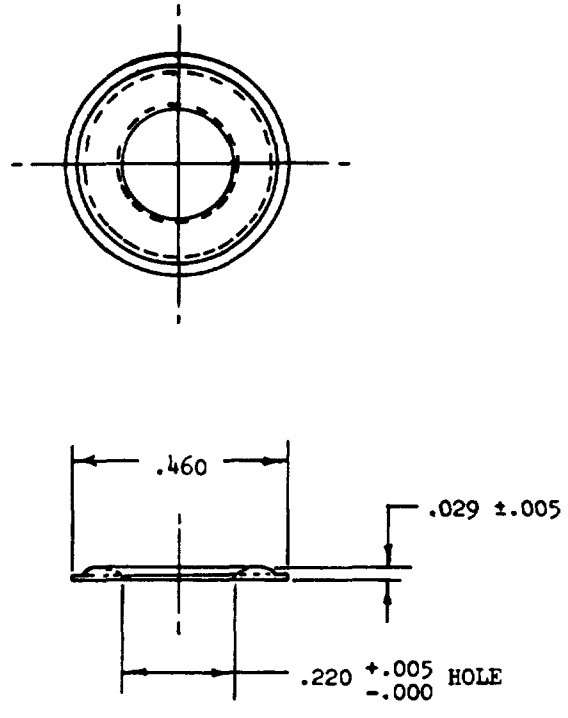
3.3 Design. The hat shall be a stiff crown model with a hatband and a wide brim. Both the crown and the brim shall be quilted with parallel lines of stitches. Each side of the crown shall have ventilators.

3.4 Construction. The hat shall be constructed in accordance with Table I, but the manufacturer is not required to follow the exact sequence of operations as listed therein.

3.4.1 Stitches, seams, and stitchings. Stitch, seam, and stitching types specified in Table I shall be as specified in FED-STD-751. Whenever two or more methods, seams, or stitch types are specified for the same part of an operation, any one of them may be used. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the hat.

3.4.1.1 Stitches per inch. The number of stitches per inch specified in Table I shall be considered the minimum and the maximum.

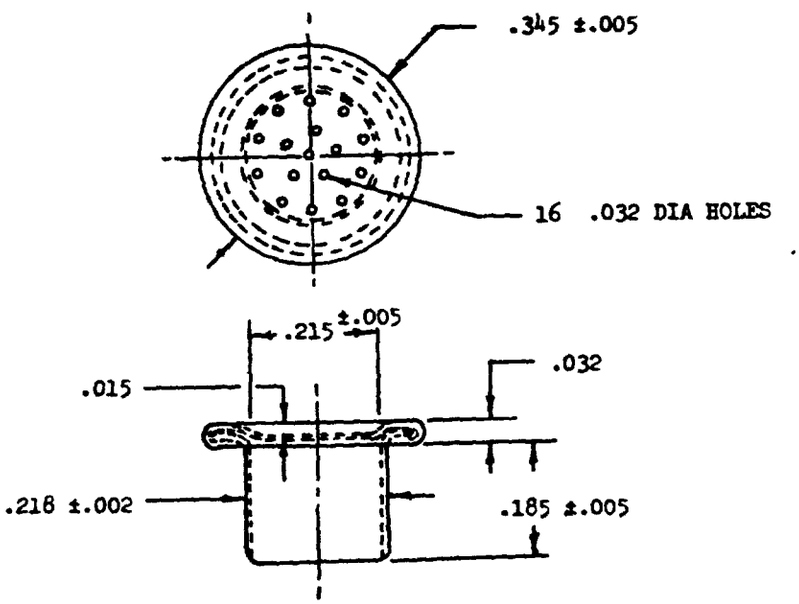
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DIMENSIONS IN INCHES

FIGURE 1. Washers

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DIMENSIONS IN INCHES

FIGURE 2. Ventilators

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TABLE I - SEWING OPERATIONS

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
1.	Cutting					
	a. Cut all materials in strict accordance with the patterns furnished, which show size, directional lines, and notches for proper assembly of parts.					
	b. Cut the hatband and the top and the bottom trims from one piece of basic material. The sweatband and the crown binding only may be cut from ends of rolls. Parts cut from ends shall approximate the shade of the basic material.					
	c. Cut the top crown and side crown from a quilted section of the one piece:					
	(1) Superimpose the oxford material on the interlining cloth, quilt together on a multineedle machine, in a straight quilting pattern, with the rows of stitching 1/4 inch apart and with the stitching in the warp direction. The loops of the stitch shall be on the mesh nylon cloth side.	101 or 301 or 401	CSv-1	8-10	P B B	 B A
	(2) Cut the top crown and the side crown pieces, in the warp direction, from the quilted material.					
	d. Cut the binding on the bias 1 to 1-1/8 inches wide. The binding for the crown seams may be cut and placed on rolls for machine application. The binding for the trim shall have one end cut with selvage and shall be cut in the lay.					
	e. Cut the hatband 2-1/8 to 2-1/4 inches wide and 26 (plus or minus 1/2) inches long on the bias.					

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TABLE I - SEWING OPERATIONS (Cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
	f. Cut the sweatband 2-1/2 (plus or minus 1/8) inches wide.					
	g. Cut the laces for chin strap 30 (plus or minus 1) inches long. Fuse the cut ends of the laces.					
	h. Cut two pieces of elastic braid for the top and the bottom hems of the headnet 22 (plus or minus 1/2) inches long. Cut two pieces of the elastic braid for loops, 6 (plus or minus 1/4) inches long					
	i. Cut two plies of brim interlining from the interlining cloth.					
2.	Replacement of damaged parts					
	a. Replace at the time of cutting any part containing holes or weakening defects such as smashes, multiple floats, or loose slubs.					
	b. Replace any part damaged during the manufacturing process by needle chews, cuts, tears, holes, mends, burns, and exposed drill holes.					
	Note: Broken or skipped stitches of the quilting stitches of the crown parts shall be repaired prior to assembly.					
3.	Shade marking. Mark, ticket, or bundle all cut parts (except parts cut from ends) to insure a uniform shade and size throughout the hat.					
4.	Baste crown parts. Baste stitch around the sides of the top crown and the side crown pieces 1/8 to 3/16 inch from the edges.	101 or 301 or 401	SSa-1	6-8	B B B	 B A

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TABLE I - SEWING OPERATIONS (Cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
5.	Make chin strap keeper					
	a. Cut leather piece for chin strap keeper, 3/4 (plus or minus 1/8) inch long by 1/2 (plus or minus 1/8) inch wide.					
	b. Punch two holes, equally spaced, in the keeper to accommodate the chin strap. The holes shall be of sufficient length and width to insure proper functioning of the keeper.					
	c. Thread the chin strap through the holes in the keeper.					
6.	Make pocket for headnet. Turn in the front and the back edges of the pocket, 3/8 to 1/2 inch, and stitch 3/16 to 1/4 inch from the folded edges forming hems.	301	LTa-1	10-14	A	A
	Note: The pockets may be cut in strips and the sides hemmed and then cut to pattern width.					
7.	Make brim. The brim shall be made of two plies of the basic material, interlined with two plies of interlining cloth, and quilted with parallel lines of stitching. The outside edge of brim shall be bound and shall finish smooth without twists or puckers.					
	a. Superimpose both plies of the brim interlining with the notches matching and bind the outside edge with plastic film with the ends of binding overlapped not less than 3/8 inch.	101 or 301	BFa-1	6-8	B B B	
	b. Superimpose the outside edges of bottom brim and the brim interlining with the notches matching, and stitch together 1/16 to 1/8 inch from the outside edges.	101 or 301	SSa-1	6-8	B B	B
	c. Superimpose the outside edges of the top brim and the brim interlining with the notches matching, and stitch together 1/16 to 1/8 inch from outside edge.	101 or 301	SSa-1	6-8	B B	B

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TABLE I - SEWING OPERATIONS (Cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
	d. In lieu of operation No. 7.b. and 7.c., superimpose the outside edges of the bottom brim, the brim interlining, and the top brim with the notches matching, and stitch together 1/16 to 1/8 inch from the outside edges in one operation.	101 or 301	SSa-1	6-8	B B	 B
	e. Single or double stitch the brim plies together with the row of stitches starting from the outside edge of the brim and continue stitching for the entire brim width. The rows of stitching shall be spaced 3/16 to 1/4 inch apart. The needle thread shall finish on the top brim in the finished hat.	301		8-10	B	B
	f. Superimpose the edges of the binding on the underside of the brim, and stitch the binding to the outside edge of the brim 1/8 to 3/16 inch from edge, holding the binding tight. The binding shall start and finish at center back (plus or minus 1/2 inch) with the selvage end of the binding on the outside of the finished brim. The ends of the binding shall lap not less than 1/2 inch.	301 or 401	BSg-2(a)	10-14	B B	B A
	g. Turn the binding over the edge of the brim, turn in raw edge and stitch 1/16 to 3/32 inch from the edge through the binding and the brim. The binding shall finish 1/4 to 3/8 inch wide on the top side of the brim.	301	BSg-2(b)	10-14	B	B
8.	Make crown. The crown shall be made of one ply of the basic material and lined with interlining cloth. The outside ply of the lining shall be quilted together prior to assembly (see operation No. 1.c.). The crow shall finish smooth and flat without twists, pleats, or puckers.					
	a. Mark (do not notch) the center front of the top crown and the side crown pieces.					

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TABLE I - SEWING OPERATIONS (Cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
	b. Join the top crown to the side crown, as indicated by the marks, with the stitching 3/16 to 1/4 inch from the edges.	101 or 301 or 401	SSa-1	10-14	B B B	 B A
	c. Bind the crown joining seam allowance with the raw edges of the binding turned in and caught in the stitching and catching the raw edges of the net pocket in the stitching (may be performed during operation No. 4.), positioned as indicated by the marks on the pattern. The binding shall finish 1/4 to 3/8 inch each side of the seam with the ends of binding extending not less than 1 inch below the bottom edge of the crown.	301 or 401	BSc-1	10-14	B B	B
	d. Baste stitch around the bottom edge of the crown 1/8 to 3/16 inch from the edge, turning the bound seam allowance of the top crown toward the center back.	101 or 301	SSa-1	6-8	B B	 B
	e. Turn the crown right side out and trim the ends of the binding even with the bottom edge of the crown.					
9.	Attach hatband to crown. The hatband shall finish 1-1/8 (plus or minus 1/16) inches wide on the outside of the hat and shall be smooth, without twists, pleats, or puckers.					
	Note: Hold the crown taut in performing all hatband stitching operations.					
	a. Fold the square end of the hatband approximately three-fourths the band width to form a triangle and position the top edge on the crown, starting at the center on the left side of the hat. Stitch the hatband to the crown 1/4 inch from the edge of the hatband, overlapping the ends not less than 1/2 inch at the top edge of the finished hatband.	101 or 301 or 401	LSs-2(a)	10-14	B B B	 B A

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TABLE I - SEWING OPERATIONS (Cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
	b. Turn the hatband down, fold the raw edge over the bottom edge of the crown and stitch 1/8 to 3/16 inch from the folded edge through all plies.	301 or 401	OSf-1	10-14	B B	B A
	c. Trim the hatband, if necessary, to have the edge even at the overlapped ends on the inside.					
	d. Single stitch the hatband with the starting rows of stitches 3/16 to 1/4 inch from the top edge of the hatband starting near or adjacent to the right back seam; continue stitching for the entire width of the hatband, with the rows of stitching 3/16 to 1/4 inch apart.	301 or 401	SSv-1	8-10	B B	B A
	Note: The starting and the ending overlap shall be not less than 1 inch.					
10.	Join brim to crown. Join the crown to the brim, matching the brim notch with the center back of the crown, and stitch 1/16 to 1/8 inch from the bottom edge of the hatband through the brim. The inside edge of the brim interlining shall be the guide in positioning the crown on the brim.	301 or 401	LSa-1	10-14	B B	B A
	Note: The brim may be basted to the crown prior to joining the crown to the brim.	101	LSa-1	4-6	B	
11.	Attach chin strap. Stitch, tack, or bartack the ends of the chin strap at each side to the brim seam allowance the full width of the strap.	301 or bartack		10-14 21-28 per bartack	B B	B B
12.	Make and attach sweatband. The seam of the sweatband shall be at the center back (plus or minus 1/2 inch) of the hat. The sweatband shall finish 1 (plus or minus 1/8) inch wide. The finished length of the sweatband shall					

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TABLE I - SEWING OPERATIONS (Cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
	conform to the measurement specified herein for the inside hat measurement that is taken along the bottom edge of the sweatband. The sweatband shall be smooth and flat without distortion and shall not be visible when the hat is worn.					
	a. Cut the sweatband to size, and overedge stitch the top edge of the sweatband.	502 or 503	EFd-1	8-10	A	A
	b. Fold the bottom edge 1 (plus or minus 1/8) inch, fold the top (overedge stitched) edge over the raw edge, and stitch through all three plies 1/16 to 1/8 inch from the edge catching the combination identification-size label in the stitching on the underside of the sweatband 1-1/2 to 2 inches to the left of the center back seam. The size shall be visible above the edge of the sweatband.	301		10-14	B	B
	c. Join the end of the sweatband with the stitching 3/16 to 1/4 inch from the edge and open seam.	301	SSa-1	10-14	B	B
	d. Turn the hat inside out.					
	e. Lap the bottom edge of the sweatband on the seam allowance of the brim. Stitch the sweatband through the crown with the stitching 1/32 to 1/16 inch from the edge of the sweatband, catching the chin strap in the stitching at both sides of the hat, and continue stitching around sweatband a second time around.	301	LSa-1	10-14	B	B
	f. Turn the hat outside out.					
13.	Blocking. Steam block the hats on proper sized pressing plugs and metal dies, shaping the hat.					

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TABLE I - SEWING OPERATIONS (Cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
14.	Attach ventilators and snap fasteners. Pre-punch holes and attach three ventilators through each side of the crown, spaced triangularly, as indicated by the marks on the pattern, with the washer portion on the inside of the hat. Install stud on wearer's right side of crown 1/2 inch below the lower ventilators directly in line with the upper ventilator. Install socket on wearer's right brim approximately 1 inch from outer edge of brim directly in line with the socket when the brim is flush against the crown.					
15.	Make headnet					
	a. Join the back seam of the headnet, with the stitching 1/4 to 3/8 inch from the edge and turn the headnet.	301	SSa-1	10-14	A	A
	b. Tie square knots in the ends of elastic cords for the top and the bottom of the headnet.					
	c. Turn under the top and the bottom of the headnet 3/8 to 5/8 inch with the elastic braid inserted. Stitch 1/4 to 3/8 inch from the edge forming hem, leaving a 3/8-inch to 1/2-inch opening at the back of the bottom hem only, not less than 1/2 inch from the center back joining seam.	301	EFa-1	10-14	A	A
	d. Fold the loops in half, knot the ends and tack on the inside through the bottom hem and the elastic braid inserted in the hem with the tacking immediately above the knot. The loops shall be spaced 6 (plus or minus 1/4) inches from the center front of the headnet with fullness distributed evenly around the headnet.	101 or 301		7-11 per tack	B B	 B

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TABLE I - SEWING OPERATIONS (Cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
	e. In lieu of the preceding operation, the loops may be caught in the hem stitching positioned as specified in operation No. 15.d.					
	f. Fold headnet to approximately 6-inch width, roll and insert in the headnet pocket in the crown of the hat.					
16.	Measure and clean hats					
	a. Measure the hat for size in accordance with the finished measurements specified herein.					
	b. Trim all ends of thread.					
	c. Remove all loose threads, spots, stains, and shade tickets.					

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3.4.1.2 Thread breaks and ends of seams. Ends of all seams and stitchings, produced with stitch type 301, when not caught in other seams or stitching shall be backtacked not less than 1/4 inch except that ends of a continuous line of stitching (all stitch types) shall overlap not less than 2 inches and shall not be backtacked. Thread breaks (all stitch types) shall be secured by stitching beyond each end of the break not less than 1/2 inch. Skipped stitches or thread breaks (stitch type 101 or 401) may be repaired using stitch type 301.

3.4.2 Marking. The cut parts of the hat shall be marked to insure a uniform shade and size throughout the hat. Any method of marking may be used except metal fastening devices, sew-on type tickets, and adhesive type tickets which will show discoloration or if the adhesive mass will adhere to the material upon removal of the tickets.

3.5 Patterns. The dies or patterns for cutting all parts of the hat shall be furnished by the manufacturer. The manufacturer's working patterns shall be identical in size and shape to the patterns referenced on Drawing 66K1622. The working patterns shall be duplicates of the Government patterns. The patterns shall provide an allowance of 1/4 inch for all seams, except for attaching the crown to the brim. The allowance for attaching the crown to the brim shall be 1/2 inch.

3.5.1 Pattern parts. The material and the number of parts cut shall be as specified in Table II.

3.6 Finished measurements. The finished measurements of the hat shall be as specified in Table III

3.7 Identification of product. Each hat shall have a combination identification-size label conforming to type I, class 4 of DDD-L-20. The label shall measure not less than 1 inch by 1-1/2 inches. The end item is intended to be laundered.

3.8 Workmanship. The finished hat shall be constructed in a thoroughly workmanlike manner and shall be free of any defects that might adversely affect serviceability.

TABLE II - PATTERN PARTS

Material	Pattern Part	Quantity Cut
Cloth, oxford, cotton warp and nylon, quarpel treated	brim - top	1
	brim - bottom	1
Cloth, oxford, cotton warp and nylon quarpel treated quilted to cloth, plain weave, nylon	top crown	1
	side crown	1
Cloth, plain weave, nylon	brim interlining	2
Cloth, netting, nylon, tulle, tricot	headnet	1
	pocket for headnet	1

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TABLE III - FINISHED MEASUREMENTS (IN INCHES)

Hat Size	Measurements 1/		
	Inside Hat (Head Measurement) (±1/4)	Crown Height at Center Front (± 3/8)	Brim Width at Center Front (±3/8)
6-3/8	20-1/8	4	3
6-1/2	20-1/2	4	3
6-5/8	20-7/8	4	3
6-3/4	21-1/4	4	3
6-7/8	21-5/8	4	3
7	22	4	3
7-1/8	22-3/8	4	3
7-1/4	22-3/4	4	3
7-3/8	23-1/8	4	3
7-1/2	23-1/2	4	3
7-5/8	23-7/8	4	3
7-3/4	24-1/4	4	3
7-7/8	24-5/8	4	3

1/ NOTES.
The finished measurement of the inside hat (head measurement) shall be taken along the bottom edge of the sweatband.
The finished measurement of the crown height at the center front shall be taken from the base of the crown to the crown seam on the outside.
The finished measurement of the brim width at the center front shall be taken from the base of the crown to the outside edge of the brim on the top side.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase orders, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105 unless otherwise specified herein.

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4.2.1 Testing of components. Except that this specificaiton shall govern in the event of conflict, components shall be tested as specified in Table IV and subsidiary specifications to the extent applicable. All requirements shall be applicable to the sample unit. Unless otherwise specified in subsidiary specifications, the lot size and sample size shall be as specified in Table V. The lot shall not be acceptable if one or more sample units fail to meet any test requirement specified. The lot size and the sample unit for testing components shall be in accordance with applicable subsidiary specifications and as follows:

Component	Lot Size Expressed as	Sample Unit for Testing
Nylon elastic braid	yards	3 continuous yards
Ventilator	gross	20 each

TABLE IV - Testing of Components

Component and Characteristic	Test Method 1/
Nylon elastic braid (see 3.2.4)	
Material identification	2/
Width	5020 of FED-STD-191
Size of nylon	2/
Number of carriers	count
Number of picks	count
Weave	visual
Number of strands of rubber	count
Gage of rubber 3/	2/
Color	visual
Colorfastness to laundering	5610 of FED-STD-191
Colorfastness to light	5660 of FED-STD-191
Ventilators (see 3.2.9)	
Diameter of head	gage
Outside diameter of barrel	gage
Length of barrel	gage
Outside diameter of washer	gage
Hole diameter of washer	gage

1/ Unless otherwise specified in the test method, one determination shall be made per sample unit.
2/ A supplier's certificate of compliance will be acceptable.
3/ Measure to the nearest 0.001 inch or number of strands side by side, actual count, prior to fabrication into finished braid.

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TABLE V - LOT SIZE AND SAMPLE SIZE

Lot Size	Sample Size
800 or less	2
801 to 22,000	3
22,001 and over	5

4.2.2 Examination of the end item. The end item shall be examined for defects. Except for defects with an asterisk (*) in the classification column, the defects found during this examination shall be classified as specified in 4.2.2.1 and 4.2.2.2. Defects with an asterisk in the classification column shall be classified as a major defect when seriously affecting appearance or serviceability and as a minor A defect when affecting appearance or serviceability but not seriously. The inspection levels shall be II of MIL-STD-105 for 4.2.2.1 and S-3 of MIL-STD-105 for 4.2.2.2. The acceptable quality level for 4.2.2.1 shall be 2.5 defects per 100 units (dhu) for major defects, 10.0 dhu for major and minor A defects, and 15.0 dhu for major and minor A and B (combined) defects. The acceptable quality level for 4.2.2.2 shall be 4.0 dhu for defects (one class).

* 4.2.2.1 Classification of defects. Defects shall be classified as follows.

	Classification		
		Minor	
	Major	A	B
BLOCKING			
a. Omitted	X		
b. Poorly blocked causing distortion of crown or hat not symmetrical affecting appearance		X	
CLEANNESS			
a. Any spot or stain on outside		X	
b. Thread end not trimmed or loose threads or one or more shade tickets not removed			X
c. Shade stamp marking (when used) exposed on outside		X	

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	Classification		
		Minor	
	Major	A	B
COMPONENTS AND ASSEMBLY			
a. Any component part or required operation omitted (unless otherwise classified herein)			
CUTTING			
a. Any component part not cut in accordance with directional lines indicated on pattern or specified requirements	X		
VENTILATORS			
a. One or more omitted or added	X		
b. Chipped, broken, or bent	X		
c. One or more improperly clinched, i.e., loose or raw edges exposed around edges of ventilator on outside	X		
d. One or more not properly clinched, i.e., raw edges of lining exposed around edges of ventilator on inside			X
e. Crown puckered or pleated in clinching one or more ventilators affecting appearance	X		
f. Not triangularly spaced	X		
g. One or more ventilators with washer on outside	X		
SNAP FASTENER			
a. Missing, mismatched or not securely clinched, or otherwise defective affecting function	X		
b. Clinched too tightly, i.e., cutting surrounding fabric	X		
c. Rough or sharp edges on stud or socket		X	
d. Stud and socket out of alignment causing noticeable twist affecting the uniformity of the brim roll front and back when closed.		X	
LABEL			
a. Missing, incorrect, or illegible	X		
b. Not securely attached		X	
c. Misplaced, i.e., nearest edge less than 1-1/2 inches or more than 2 inches from sweatband joining seam			X

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	Classification		
		Minor	
	Major	A	B
MATERIAL DEFECTS AND WORKMANSHIP DAMAGES (Material defects and workmanship damaged shall be classified as defects only when the condition is one which weakens the fabric or hat or when it is conspicuously located so that it is clearly noticeable. Nonweakening conditions which are inconspicuous shall not be classified as defects.)			
a. Any hole		*	
b. Weakening defect such as smash, multiple float, shade bar or slub		*	
c. Cut, tear, rip, mend or burn	X		
d. Needle chew		*	
SEAMS AND STITCHING			
a. General			
(1) Seam twisted, puckered, or pleated, affecting appearance		X	
(2) Part of hat or headnet caught in any unrelated operation or stitching		*	
(3) Thread(s) used on outside not same shade or not satisfactorily matching shade of hat		*	
(4) End of seam (except continuous line of stitching) when not caught in other seam or stitching backtacked less than 1/4 inch		X	
(5) End of a continuous line of stitching overlapped less than 2 inches unless otherwise specified		X	
(6) Thread break (all stitch types) stitched less than 1/2 inch beyond each end of break			X
b. Gage of stitching (to be scored when condition exists on major portion of seam, i.e., over one-half of seam)			
(1) Irregular, i.e., unevenly gaged, or stitchings not uniform, affecting appearance		X	
(2) Beyond range or width specified or varies more than 1/16 inch when no range is specified			X
(3) Edge or raised stitching sewn too close to edge resulting in damage to cloth		*	

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	Classification		
		Minor	
	Major	A	B
c. Open seams (One or more broken or two or more skipped or run-off stitches shall constitute an open seam.)			
(1) Any seam open on outside			
-up to 1/4 inch inclusive		X	
-more than 1/4 inch	X		
(2) Any seam open on inside of hat or headnet			
-up to 1/4 inch inclusive			X
-more than 1/4 inch to 1 inch inclusive		X	
-more than 1 inch	X		
d. Raw edges (Raw edges, except on headnet, not securely caught in stitching shall be classified as an open seam.)			
(1) Any raw edge on outside of hat (excluding headnet)			
-1/4 to 1/2 inch inclusive		X	
-more than 1/2 inch	X		
(2) Any raw edge on inside of hat more than 1/4 inch		X	
e. Runoffs (Runoffs on joining seams, when resulting in an open seam, shall be classified as an open seam.)			
(1) Runoffs on binding stitching			
-1/4 inch to 1/2 inch inclusive		X	
-over 1/2 inch	X		
f. Seam and stitch types			
(1) Not specified seam or stitch type		X	
(2) Looper thread on outside (when stitch type 401 is used)	X		
(3) Chain of stitch type 101 on outside of hat	X		
g. Stitch tension (Puckering is evidence of tight tension. When puckering is evident, seam shall be tested by exerting pull in lengthwise direction of seam or stitching.)			
(1) Loose tension resulting in a loose seam		X	
(2) Loose tension on top quilting stitches or edge stitching, resulting in loosely exposed lower thread or top thread		X	

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	Classification		
		Minor	
	Major	A	B
(3) Tight tension (stitches breaking when normal strain is applied to the lengthwise direction of seam or stitching)	X		
h. Stitches per inch (to be scored only when the condition exists on major portion of seam on stitching)			
(1) Applicable to 6-8, 8-10, and 10-14 stitches per inch, excluding overedge stitches			
-one stitch less than minimum specified			X
-two stitches less than minimum specified		X	
-three or more stitches less than minimum specified	X		
-more than maximum specified			X
(2) Applicable to 8-10 stitches per inch overedge stitches			
-one stitch less than minimum specified			X
-two stitches less than minimum specified		X	
-more than maximum specified			X
SHADED PARTS			
a. Badly shaded parts (except those cut from ends)	X		
b. Parts permitted to be cut from ends badly shaded except sweatband			X
BRIM			
a. Twisted, stretched, or poorly shaped	X		
b. Rows of quilting stitches less than 3/16 inch or more than 1/4 inch apart		X	
c. Needle thread of quilting stitches finished on underside of brim			X
d. Fullness between rows of quilting stitches on underside of brim		X	
e. Binding twisted, puckered, or pleated	X		
f. End of binding not selvage or not within 1/2 inch of center back of brim			X
g. Binding finished less than 1/4 inch or more than 3/8 inch wide on top of brim for a distance of 2 inches or more		X	

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	Classification		
		Minor	
	Major	A	B
h. Irregular in width			
-more than 1/4 to 1/2 inch inclusive		X	
-more than 1/2 inch	X		
i. Binding reversed		X	
CROWN			
a. Crown puckered or pleated at joining of top crown to side crown, affecting appearance		X	
b. Poorly shaped, twisted, stretched, or pleated			
c. Puckered or pleated at joining of side crown to brim		X	
d. Headnet pocket missing	X		
e. Front or back edge of headnet pocket not hemmed			X
f. Outside crown and lining not quilt stitched together	X		
g. One or both sides of headnet pocket not completely caught in crown binding seam		X	
h. Binding omitted from crown joining seam	X		
i. Width of binding less than 1/4 inch or more than 5/16 inch wide			X
j. End of binding not trimmed even with edge of crown			X
k. Seam allowance of crown seam not toward center back		X	
HATBAND			
a. Twisted or not uniformly shaped	X		
b. Finished width less than 1-1/6 inches or more than 1-3/16 inches wide on outside of hat		X	
c. End of hatband not folded under	X		
d. Rows of stitches less than 3/16 inch or more than 1/4 inch apart		X	
e. Fullness between rows of stitching			X
f. Top end of hatband offcenter of left side by 3/4 inch or more			X

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	Classification		
		Minor	
	Major	A	B
g. Starting row of stitching not adjacent to right back seam		X	
h. Starting and ending overlap of hatband stitching less than 1 inch		X	
CHIN STRAP AND KEEPER			
a. Chin strap missing	X		
b. End not tacked or bartacked to brim seam allowance		X	
c. End of chin strap not fused		X	
d. Keeper missing	X		
e. Keeper less than 5/8 inch or more than 7/8 inch long or less than 3/8 inch or more than 5/8 inch wide			X
f. Holes in keeper too small or too large not permitting chin strap to function properly, i.e., falls freely from set position or keeper immovable	X		
g. Chin strap fused over 1/4 inch in length	X		
SWEATBAND			
a. Twisted or not flat against crown		X	
b. Exposed 1/8 inch or more below hat for a distance -1/2 inch to 3 inches inclusive -over 3 inches		 X	X
c. Width less than 7/8 inch or more than 1-1/8 inches		X	
d. Joining seam of sweatband more than 1/2 inch from center back of hat		X	
HEADNET			
a. Headnet missing	X		
b. One or both elastic cords omitted	X		
c. Loop not finishing on inside or not facing outward			X
d. One or more elastic braids not knotted or not specified type of knot			X
e. One or both hems less than 3/8 inch or more than 5/8 inch wide		X	

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	Classification		
		Minor	
	Major	A	B
f. Opening in bottom hem omitted		X	
g. Opening in bottom hem measuring less than 3/8 inch or more than 1/2 inch long			X
h. One or both loops omitted or not attached to bottom hem	X		
i. Ends of one or both loops not tied prior to tacking, not tacked through braid in hem, or not caught in hem stitching		X	
j. Loops not on front of headnet		X	
k. Headnet not folded and inserted in pocket			X

4.2.2.2 Finished measurements. Any finished measurement deviating from the finished measurements specified in Table III shall be classified as a finished measurement defect.

* 4.3 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each hat shall either have the crown stuffed with tissue paper which shall immobilize the chin strap and permit the hat to be nested approximately 2-1/2 inches or shall be fitted with a commercial type hat ring which will permit nesting.

5.1.2 Level C. Hats shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Twelve hats of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to class weather-resistant, grade V3c, style FOL of PPP-B-636. The inside dimensions of the shipping container shall approximate 27-3/4 inches in length, 15-1/2 inches in width, and 16 inches in depth. The inside of each shipping container shall be fitted with a one-piece, full-depth, style S, A-flute, double-faced corrugated fiberboard liner having a minimum bursting strength of 200 pounds per square inch (psi) conforming to class domestic of PPP-B-636. The liner shall divide the container length into two equal size cells

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measuring inside approximately 13-1/2 inches in length and 15 inches in width. Approximate dimensions for the shipping container and the liner are furnished as a guide only. The hats shall be positioned in two stacks of six nested hats per stack. The top and the bottom hats in each stack shall have a double-faced corrugated fiberboard die-cut pad fitting around the crown of the hat and resting evenly on the brim. The pad shall measure 1/4 inch less than the inside length and width dimensions of the cell and shall be notched at one corner to permit easy removal. The fiberboard material used for the pad shall be the same as that specified for the liner. Each container shall have the contents completely covered on the top and the bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. All seams, corners, and manufacturer's joint of the fiberboard containers shall be sealed by the application of 3-inch-minimum-width olive-drab-color cloth tape conforming to type IV, class 1 of PPP-T-60. Each shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. Toward the end of the contract or when there are less than the required amount of the same size, mixed sizes (see 5.3.1) may be packed in the same shipping container.

5.2.2 Level B. Twelve hats of one size only, packaged as specified in 5.1, shall be packed flat in a fiberboard shipping container conforming to type CF, class domestic, style OSC (3-inch-minimum overlap) of PPP-B-636. The minimum bursting strength of the fiberboard shall be 200 psi. The inside dimensions of the shipping container shall approximate 27-3/4 inches in length, 15-1/2 inches in width, and a 16 inches in depth. The inside of each shipping container shall be fitted with a one-piece, full depth, style S, A-flute, double-faced corrugated fiberboard liner having a minimum bursting strength of 200 psi, conforming to class domestic of PPP-B-636. The liner shall divide the container length into two equal size cells measuring inside approximately 13-1/2 inches in length and 15 inches in width. Approximate dimensions for the shipping container and the liner are furnished as a guide only. The hats shall be positioned in two stacks of six nested hats per stack. The top and the bottom hats in each stack shall be provided with a double-faced corrugated fiberboard die-cut pad fitting around the crown of the hat and resting evenly on the brim. The pad shall measure 1/4 inch less than the inside length and width dimensions of the cell and shall be notched at one corner to permit easy removal. The fiberboard material used for the pad shall be the same as that specified for the shipping container and the liner. Each container shall have the contents completely covered on the top and the bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. All flaps of the fiberboard shipping container shall be securely sealed, with an adhesive commercially used for the specified product being packed by applying throughout the entire area of contact between the flaps or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive. Toward the end of the contract or when there are less than the required amount of the same size, mixed sizes (see 5.3.1) may be packed in the same shipping container.

5.2.3 Level C. Hats, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

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5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stencilling, a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and sizes contained therein.

6. NOTES

6.1 Intended use. The hat is intended for use by male military personnel for special warfare purposes in tropical and semitropical areas.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.1).

c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.3 Sample and shade samples. For access to sample of hat and shade samples of materials, address the procuring activity issuing the invitation for bids.

6.4 Figure. Figure 3 shows general design and is furnished for information only.

6.5 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

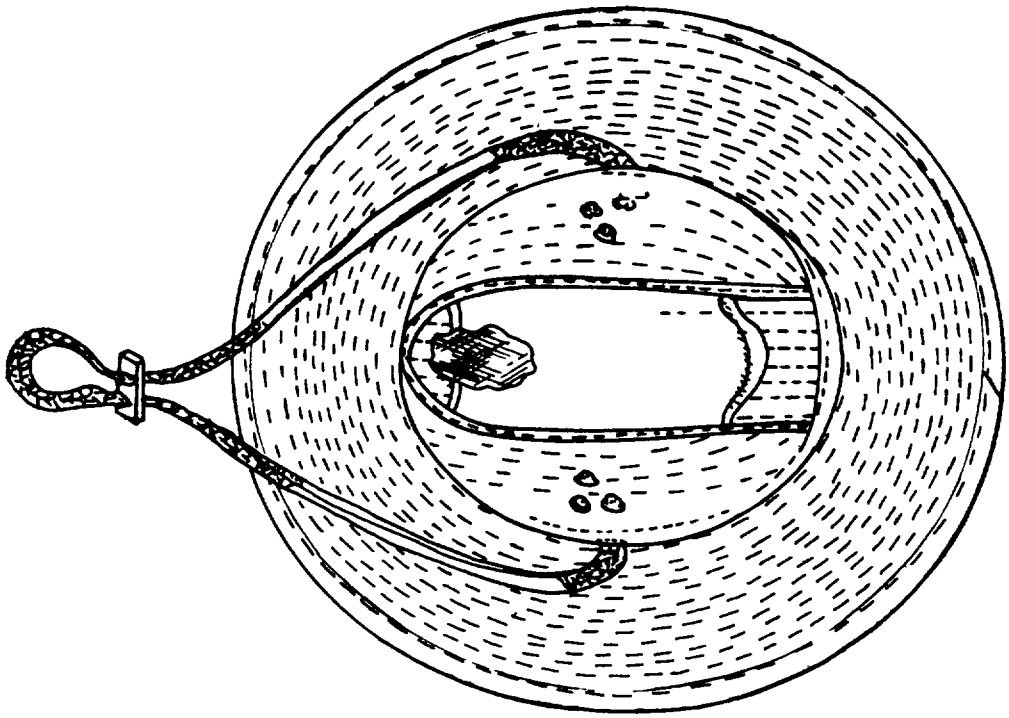
Custodian.
Air Force - 82

Review Activities.
Air Force - 11, 82

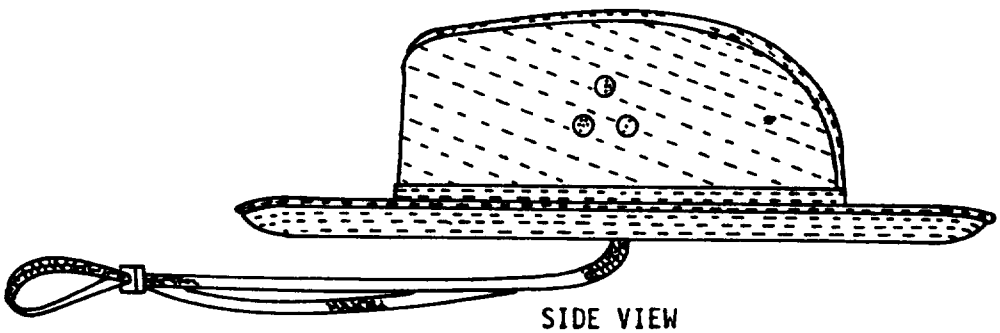
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INSIDE VIEW



SIDE VIEW

FIGURE 3. Hat and Insect Net HGU-18/P

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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